THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Light travels in: 1

(a) circles **(b)** straight lines **(c)** curved lines **(d)** zigzag lines

2. For a shadow to form, we need: 1

(a) a source only
(b) a screen only
(c) an object only
(d) a source of light, an opaque object and a screen

3. The bouncing back of light from a surface is: 1

(a) dispersion **(b)** reflection **(c)** absorption **(d)** refraction

4. The image formed by a plane mirror is: 1

(a) real and inverted **(b)** virtual and inverted
(c) real and erect **(d)** virtual and erect

5. Objects that give out their own light are: 1

(a) luminous objects **(b)** non-luminous objects
(c) opaque objects only **(d)** transparent objects

6. The size of a shadow becomes larger when the object is moved: 1

(a) behind the screen **(b)** closer to the screen
(c) closer to the light source **(d)** farther from the source

7. A surface that reflects light well is: 1

(a) a rough stone **(b)** a black cloth **(c)** a mirror **(d)** soil

8. The size of the image in a plane mirror is: 1

(a) same as the object **(b)** double
(c) smaller **(d)** larger

9. Which of the following is a luminous object? 1

(a) a mirror **(b)** the Sun **(c)** the Moon **(d)** a book

10. The darkest part of a shadow is the: 1

(a) penumbra **(b)** halo **(c)** umbra **(d)** image

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the difference between luminous and non-luminous objects? 2

12. What is a pinhole camera? 2

13. What is a periscope used for? 2

14. What are transparent, translucent and opaque materials? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain with an activity that light travels in straight lines. 3

16. A person stands 2 m in front of a plane mirror. Find the distance of the image from the mirror and between the person and the image. 3

17. Explain the formation of umbra and penumbra. 3

OR

Describe how a shadow changes in size when an object is moved towards the screen. 3

18. Explain how a periscope works. 3

OR

Why does the Moon not produce its own light? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the activity to show that light travels in a straight line and to produce a shadow, and discuss the factors affecting the size of a shadow. 5

OR

Describe a kaleidoscope: how it is made and the pattern formed. 5

20. Describe the working of a pinhole camera. Explain why the image is inverted. 5

OR

Discuss luminous and non-luminous objects with five examples and how we see non-luminous objects. 5

END OF QUESTION PAPER — Total: 40 Marks

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